

Notes: Alok & Ayyagari (RFS, 2020)

Politics, State Ownership, and Corporate Investments

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1 Overview

This paper asks whether politicians *use state-owned enterprises (SOEs)* to manipulate the timing and composition of corporate investment for electoral gain. Using Indian district-level project announcements (CAPEX) matched to election timing, the authors document a strong *political investment cycle* for SOEs: SOE project announcements and spending rise in election years, especially in politically competitive districts, and shift toward *visible* projects (construction, community development). Private firms do *not* display a comparable cycle. These politically timed SOE investments help incumbents electorally, but markets react negatively to election-year SOE projects (for partially privatized SOEs), and the usual positive association between SOE and private investment weakens in election years (crowding-out/attenuation).

2 Research question and hypotheses

2.1 Core question

Let I denote corporate investment (project announcements, costs, implementation). The core question is:

Does political control over SOEs distort I around elections, and with what consequences?

2.2 Key hypotheses (as testable predictions)

1. **Political investment cycle (PIC):** SOE investment activity increases in election years; private firms do not show the same systematic cycle.
2. **Tactical redistribution/swing targeting:** the SOE election-year spike is stronger in *close* (politically competitive) districts than in safe districts.
3. **Visibility/composition channel:** election-year SOE projects shift toward *visible* categories that voters can observe (e.g., construction).
4. **Electoral returns:** election-year SOE investments (especially visible ones) improve incumbent electoral outcomes.
5. **Value destruction:** if election-year SOE investments are politically motivated and not value maximizing, announcement returns for partially privatized SOEs should be negative (relative to comparable private projects).
6. **Real allocation effects:** if SOE investment is distorted, it may crowd out/attenuate private investment in election years (a real economic consequence beyond mere timing).

3 Data and measurement

3.1 Main datasets

- **CAPEX project database (India):** project announcements and characteristics: ownership type, industry, project cost, implementation status, dates.
- **Election Commission of India:** timing of national and state elections; district-level electoral outcomes and margins.
- **Firm financials (Prowess) for some tests:** firm-level investment (capex), controls.
- **Policy uncertainty (Baker et al.-style index):** annual series used to study uncertainty vs. investment.
- **Stock returns:** to compute excess returns and CAPM abnormal returns around announcement/implementation dates, using S&P Nifty as market benchmark.

3.2 Units of observation

1. **District-year (main PIC tests):** i indexes state, j district, t year. Outcomes are counts/costs of projects in (i, j, t) .
2. **Project-level (composition and event studies):** p indexes a project, matched to a firm f and located in a district-year environment.
3. **Firm-year (firm financial investment tests):** i indexes firm, t year.

3.3 Key variables

I keep the paper’s notation:

- Election_{it} : indicator for an election year (state elections vary by state-time; national elections are common across districts in a given year).
- Scheduled_{it} : indicator that 5 years have elapsed since the last election (used as an instrument for state-election timing).
- Close_{ijt} : indicator that the incumbent’s previous margin of victory/loss is $< 5\%$ (political competition).
- LC_{ijt} : “less contested” (safe) districts: incumbent margin above 75th percentile.
- Y_{ijt} : investment outcome in district-year: number of projects, SOE share, $\log(1+\text{cost})$, etc.
- Visible projects: construction + community development (parks, recreation centers).

4 Empirical models

4.1 Baseline district-year investment regression (Eq. (1))

The baseline specification (used for Tables 3–4) is:

$$Y_{ijt} = \alpha_0 + \beta_1 \text{Election}_{it} + \beta_2 C_{it} + \mu_j + \mu_t + \varepsilon_{ijt}, \quad (1)$$

where:

- Y_{ijt} is a district-year investment outcome (separately for SOEs and private firms, and for shares/cost ratios).
- C_{it} includes state-level real GDP growth (and possibly other macro controls, depending on the table).
- μ_j are district fixed effects; μ_t are year fixed effects.

Interpretation. β_1 measures how investment outcomes differ in election years versus non-election years *within a district*, net of common year shocks.

4.2 Instrument for election timing in state elections (Eq. (2))

Because state elections are constitutionally scheduled every 5 years but sometimes occur early, the paper uses:

$$\text{Election}_{it} = \alpha_0 + \beta_1 \text{Scheduled}_{it} + \beta_2 C_{it} + \mu_j + \mu_t + \nu_{it}, \quad (2)$$

and then estimates eq. (1) by 2SLS, using Scheduled_{it} as an instrument for Election_{it} .

Identification intuition. The scheduled component of elections is plausibly exogenous to contemporaneous district-level investment demand shocks. This strengthens the case that an election-year jump in SOE activity is *political timing*, not local demand.

4.3 Political competition heterogeneity (Eq. (3))

To test whether manipulation is stronger in swing districts:

$$Y_{ijt} = \alpha_0 + \beta_1 \text{Election}_{it} + \beta_2 \text{Close}_{ijt} + \beta_3 (\text{Election}_{it} \times \text{Close}_{ijt}) + \gamma C_{it} + \mu_j + \mu_t + \varepsilon_{ijt}. \quad (3)$$

Interpretation. $\beta_3 > 0$ for SOE outcomes supports tactical targeting of competitive districts.

4.4 Election-cycle dynamics (Table 5 specification)

Let Sc_{it}^{-n} indicate the year relative to the next/previous *scheduled* election (the election year is the omitted category). The dynamics specification is:

$$Y_{ijt} = \alpha_0 + \sum_{n=1}^4 \beta_n Sc_{it}^{-n} + \gamma C_{it} + \mu_j + \mu_t + \varepsilon_{ijt}. \quad (4)$$

Interpretation. Negative β_n (for SOEs) means those non-election years have fewer projects than the election year (a hump in election years).

4.5 Policy uncertainty and investment (Table 6)

With a national annual policy uncertainty index PU_t :

$$Y_{jt} = \alpha_0 + \theta_0 \log(PU_t) + \theta_1 \log(PU_{t-1}) + \eta \text{Election}_t + \mu_j + \varepsilon_{jt}, \quad (5)$$

estimated separately for central SOEs and private firms (district FE; year FE is not usable when Election_t is common across districts).

4.6 Political authority: cabinet ministers (Table 7)

Let Federal_{jt} indicate a district represented by a Member of Parliament who is also a federal cabinet minister:

$$Y_{jt} = \alpha_0 + \eta \text{Election}_t + \phi \text{Federal}_{jt} + \psi (\text{Election}_t \times \text{Federal}_{jt}) + \gamma C_t + \mu_j + \varepsilon_{jt}. \quad (6)$$

4.7 Political ideology: left vs right (Table 8)

Let Left_{jt} indicate a left-wing incumbent. Then:

$$Y_{jt} = \alpha_0 + \eta \text{Election}_t + \phi \text{Left}_{jt} + \psi (\text{Election}_t \times \text{Left}_{jt}) + \gamma C_t + \mu_j + \varepsilon_{jt}. \quad (7)$$

Interpretation. $\psi > 0$ for SOE investment suggests stronger political use of SOEs under left-wing incumbents.

4.8 Project-level composition and “visibility” (Table 9, IV)

At the project level:

$$Y_{fpt} = \alpha_0 + \beta_1 \text{SOE}_f + \beta_2 \text{Election}_{pt} + \beta_3 (\text{Election}_{pt} \times \text{SOE}_f) + \mu_{\text{ind}} + \varepsilon_{fpt}, \quad (8)$$

where Y_{fpt} is a project characteristic (size, expansion, stalled/abandoned) or an industry-indicator outcome (construction, community development, etc.). For state elections, Scheduled instruments

for Election and Scheduled $\times$ SOE instruments for Election $\times$ SOE.

4.9 Electoral returns: do SOE projects help incumbents? (Table 10)

Restrict to election years and regress electoral outcomes on SOE projects:

$$\text{Outcome}_{ijt} = \alpha_0 + \beta \text{NumSOEProjs}_{ijt} + \gamma C_{it} + \mu_j + \mu_t + \varepsilon_{ijt}, \quad (9)$$

where the outcome is either: (i) number of constituencies won by the incumbent party in district j , or (ii) incumbent margin of victory. The paper further splits SOE projects into *visible* vs other.

4.10 Event-study valuation model (Eq. (6))

Return measures. Let $r_{f,\tau}$ be firm f 's stock return and $r_{m,\tau}$ the S&P Nifty return.

- **Excess return over an event window $\mathcal{W}$:**

$$\text{ER}_f(\mathcal{W}) = \sum_{\tau \in \mathcal{W}} (r_{f,\tau} - r_{m,\tau}).$$

- **Abnormal return (CAPM):** estimate CAPM over the prior 3 months:

$$r_{f,\tau} = \alpha_f + \beta_f r_{m,\tau} + u_{f,\tau},$$

then compute

$$\text{AR}_f(\mathcal{W}) = \sum_{\tau \in \mathcal{W}} (r_{f,\tau} - (\hat{\alpha}_f + \hat{\beta}_f r_{m,\tau})).$$

The cross-sectional event regression is:

$$Y_{fp} = \alpha_0 + \beta_1 \text{SOE}_f + \beta_2 Z_p + \beta_3 (Z_p \times \text{SOE}_f) + \beta_4 X_f + \mu_{\text{ind}} + \varepsilon_{fp}, \quad (10)$$

where Y_{fp} is ER or AR around the project announcement (or implementation), and Z_p is either Election (Table 11) or Close (Table 12). The key prediction is $\beta_3 < 0$ if politically driven SOE projects destroy value.

4.11 Crowding-out / attenuation of private investment (Table 13 and Table A13)

Define Y_{ijt+L} as private investment outcomes L years ahead (counts or log-cost). The specification is:

$$Y_{ijt+L} = \alpha_0 + \beta_1 \text{NumSOEProjs}_{ijt} + \beta_2 \text{Election}_{it} + \beta_3 (\text{NumSOEProjs}_{ijt} \times \text{Election}_{it}) + \gamma C_{it} + \mu_j + \mu_t + \varepsilon_{ijt}. \quad (11)$$

Interpretation. $\beta_1 > 0$ indicates “crowding in” (SOE activity co-moves positively with private investment). $\beta_3 < 0$ indicates that this association is weaker in election years (attenuation/crowding out in politically distorted periods).

5 Main results: interpretation table-by-table and figure-by-figure

5.1 Reading strategy

For each table/figure, I keep three questions in mind: (i) what is the dependent variable and sample?, (ii) what is the coefficient of interest?, and (iii) what is the economic meaning?

5.2 Table 1: summary statistics (district-year)

What it is. Table 1 reports means (and SDs) for the main district-year variables in the national-election sample and the state-election sample.

What I learn.

- Investment activity is *lumpy*: average counts per district-year are well below 2 projects, with many zeros, motivating strong use of fixed effects and careful interpretation.
- SOE project activity (both counts and costs) is smaller in level than private activity, but is systematically related to political variables (to be shown in Tables 3–4).

5.3 Table 2: project characteristics and composition (SOE vs private; election vs non-election)

What it is. Table 2 compares project-level characteristics by ownership type and by election timing.

Key reads.

- **SOE vs private (Panel A):** SOE projects take longer to implement/complete (consistent with bureaucratic process), and are less likely to be stalled/abandoned than private projects.
- **Election vs non-election within SOEs (Panel B):** election-year SOE projects are *larger* (higher log cost) and more likely to be expansions/new units.
- **Industry mix (Panel C):** SOEs tilt toward construction/community development in election years, consistent with visibility.

5.4 Table 3: baseline political investment cycle (PIC)

Model. This table is the empirical implementation of eq. (1) for national elections (Panel A) and state elections (Panels B–C), with Panel C using the IV design from eq. (2).

Table 1
Summary statistics
A. National elections

Variables	N	Mean	Median	SD
<i>Political variables</i>				
Election year	5,081	0.279	0	0.448
Absolute margin	5,081	0.144	0.104	0.131
Close	5,081	0.272	0	0.445
Less contested	5,081	0.250	0	0.434
<i>Number of projects</i>				
Central SOEs	5,081	0.283	0	0.829
State SOEs	5,081	0.475	0	1.66
Nongovernment Firms	5,081	1.865	0	6.55
All	5,081	2.623	1	7.89
<i>Cost of projects (Rs. million)</i>				
Central SOEs	4,094	269.387	0	2,844.744
State SOEs	4,094	170.468	0	1,664.909
Nongovernment firms	4,094	768.074	0	3,924.863
All	4,094	1207.929	200	5,523.350
<i>Announced dummy</i>				
Central SOEs	5,081	0.168	0	0.374
State SOEs	5,081	0.220	0	0.414
Nongovernment firms	5,081	0.389	0	0.487
All	5,081	0.503	1	0.500
B. State elections				
Variables	N	Mean	Median	SD
<i>Political variables</i>				
Election year	8,456	0.209	0	0.407
Scheduled	8,456	0.176	0	0.381
Absolute margin	8,456	0.092	0.075	0.079
Close	8,456	0.373	0	0.483
Less contested	8,456	0.251	0	0.434
<i>Number of projects</i>				
State SOEs	8,456	0.396	0	1.41
Central SOEs	8,456	0.235	0	0.762
Nongovernment Firms	8,456	1.480	0	5.67
All	8,456	2.111	0	6.825
<i>Cost of projects (Rs. million)</i>				
State SOEs	7,061	126.725	0	1,295.968
Central SOEs	7,061	225.077	0	2,385.309
Nongovernment firms	7,061	643.921	0	3,660.417
All	7,061	995.723	101.375	4,941.395
<i>Announced dummy</i>				
State SOEs	8,456	0.190	0	0.393
Central SOEs	8,456	0.138	0	0.345
Nongovernment firms	8,456	0.323	0	0.467
All	8,456	0.429	0	0.495

This table reports the summary statistics of the key variables used in our analysis. Panel A reports the summary statistics for the data on national elections. Panel B reports data on the same variables but for state elections. The data cover the period 1995–2009 and come from the election commission of India. The unit of observation is a district-year. National (state) elections happen in 435 (594) unique districts. The appendix defines all variables.

Table 2
Summary Statistics on Projects
A. SOEs vs. nongovernment

	Central and state SOEs			Nongovernment		
	(1)			(2)		Difference (3)
<i>Type of Projects</i>						
Years to completion	3.68			2.02		1.66***
Years to implementation	1.64			0.94		0.70***
Expansion/new unit	0.91			0.99		−0.08***
Ln(Cost)	3.89			4.12		−0.23***
Stalled/abandoned	0.07			0.11		−0.04***
Stalled/abandoned & Potentially abandoned	0.09			0.12		−0.03***
N	5,568			13,413		
B. Election vs. off-election years						
	Central and State SOEs			Nongovernment		
	Election (1)	Off-Election (2)	Difference (3)	Election (4)	Off-Election (5)	Difference (6)
<i>Type of projects</i>						
Years to completion	3.07	3.08	−0.01	1.86	1.87	−0.01
Years to implementation	1.65	1.65	0.00	0.80	1.00	−0.20***
Expansion/new unit	0.92	0.89	0.03**	0.98	0.99	−0.01***
Ln(Cost)	4.01	3.84	0.17***	3.88	4.24	−0.36***
Stalled/abandoned	0.07	0.07	0.00	0.12	0.11	0.01
Stalled/abandoned & Potentially abandoned	0.11	0.08	0.03***	0.12	0.11	0.01**
N	1,214	4,354		2,743	10,670	
C. Industry of projects						
	Central and state SOEs			Nongovernment		
	Election (1)	Off-election (2)	Difference (3)	Election (4)	Off-election (5)	Difference (6)
Construction	0.47	0.38	0.09***	0.11	0.1	0.01
Community development (parks and recreation centres)	0.03	0.02	0.01**	0.02	0.02	0
Education	0.02	0.04	−0.02**	0.01	0.01	0
Health and social work	0.01	0.03	−0.02***	0.02	0.02	0
Utilities (includes electricity, gas, and water)	0.18	0.18	0	0.08	0.09	−0.01*
Transport, storage and communications	0.09	0.11	−0.02*	0.02	0.02	0
Hotels and restaurants	0.01	0.02	−0.01**	0.05	0.06	−0.01
Manufacturing	0.07	0.06	0.01	0.52	0.5	0.02*
Mining	0.04	0.04	0	0.03	0.03	0
Real estate leasing	0.09	0.12	−0.03***	0.13	0.15	−0.02**
Miscellaneous	0.003	0.003	0	0.01	0.01	0

This table reports summary statistics on the characteristics of projects announced by SOEs (both state and central SOEs) and nongovernment firms. Panel A compares the characteristics of projects announced by SOEs and nongovernment firms. Panel B compares the characteristics of projects announced by SOEs and nongovernment firms during election and off-election years. * $p < .1$; ** $p < .05$; *** $p < .01$.

Panel C reports summary statistics on the fraction of projects announced by SOEs (both state and central SOEs) and nongovernment firms across eleven industry categories based on National Industry Classification codes: Construction, Community Development (parks and recreation centers), Education, Health and Social Work, Utilities (includes Electricity, Gas, and Water), Transport, Storage, and Communications, Hotels and Restaurants, Manufacturing, Mining, Real Estate Leasing, and Miscellaneous (includes Wholesale, Retail, and Financial Intermediation). * $p < .1$; ** $p < .05$; *** $p < .01$.

What the coefficients say.

- **National elections (central SOEs):** election years increase central-SOE project activity (e.g., positive coefficients on number of projects and on log total cost). Private firms do not show a similar increase.
- **State elections (state SOEs):** election years increase state-SOE investments (OLS and IV), again with little/no corresponding increase for private firms.

Economic magnitude (rule-of-thumb). Using Table 1 means, an effect like 0.109 additional SOE projects per district-year (state-elections IV) is about $0.109/0.396 \approx 27.5\%$ of mean state-SOE projects. For central SOEs in national elections, $0.048/0.283 \approx 17\%$.

Table 3
Elections and corporate investments

	Number of projects		Percentage	Announced dummy		Cost ratio	ln(Cost)
	Central SOEs (1)	Nongovt firms (2)	Central Total (3)	Central SOEs (4)	Nongovt firms (5)	Central cost Total cost (6)	Central SOEs (7)
<i>A. National elections</i>							
Election	.048*** (.018)	-.109 (.072)	.016** (.007)	.022** (.010)	.014 (.011)	.015* (.008)	.590*** (.089)
State-level real	.337** (.136)	6.229*** (.957)	-.006 (.054)	.117 (.074)	.695*** (.105)	-.010 (.065)	.180 (.304)
GDP growth	.257*** (.009)	1.619*** (.042)	.074*** (.003)	.158*** (.004)	.354*** (.006)	.082*** (.004)	.001 (.044)
Constant	5.039 (.480)	5.039 (.552)	5.039 (.185)	5.039 (.312)	5.039 (.442)	4.061 (.193)	4.061 (.302)
No. observations	5,039	5,039	5,039	5,039	5,039	4,061	4,061
R ²	.480	.552	.185	.312	.442	.193	.302
<i>B. State elections (OLS)</i>							
Election	.106** (.046)	-.086 (.053)	.032*** (.007)	.029*** (.009)	-.012 (.009)	.031*** (.008)	.103*** (.024)
State-level real	-.806*** (.172)	.334 (.297)	-.202*** (.049)	-.267*** (.062)	.100 (.078)	-.167*** (.057)	-.780*** (.183)
GDP growth	.015 (.041)	.088 (.108)	.021*** (.008)	.025** (.011)	.116*** (.015)	.012 (.008)	.019 (.029)
Constant	8.412 (.315)	8.412 (.622)	8.412 (.192)	8.412 (.324)	8.412 (.511)	7.017 (.178)	7.017 (.239)
No. observations	8,412	8,412	8,412	8,412	8,412	7,017	7,017
R ²	.315	.622	.192	.324	.511	.178	.239
<i>C. State elections (IV)</i>							
Election	.109** (.054)	.038 (.058)	.029*** (.008)	.025*** (.010)	-.012 (.010)	.030*** (.008)	.107*** (.028)
State-level real	-.807*** (.172)	.328 (.297)	-.202*** (.049)	-.267*** (.062)	.100 (.077)	-.167*** (.055)	-.781*** (.183)
GDP growth	.015 (.045)	.088 (.111)	.021*** (.008)	.025*** (.011)	.116*** (.015)	.012 (.008)	.019 (.030)
Constant	8.412 (.33,099)	8.412 (.622)	8.412 (.192)	8.412 (.324)	8.412 (.511)	7.017 (.178)	7.017 (.239)
No. observations	8,412	8,412	8,412	8,412	8,412	7,017	7,017
Wald F-stat	33,099	33,099	33,099	33,099	33,099	28,000	28,000

This table shows how investments by SOEs and private firms vary by national elections (panel A) and state elections (panels B and C). The table reports estimates from the following panel regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Election}_{it} + \beta_2 \times C_{it} + \mu_j + \mu_t + \varepsilon_{ijt},$$

Table 4
Political competition, patronage, and corporate investments

	Close				Less contested			
	Number of projects	Percentage	Cost ratio	ln(Cost)	Number of projects	Percentage	Cost ratio	ln(Cost)
	State SOEs (1)	Nongovt firms (2)	State SOEs (3)	State cost Total cost (4)	State SOEs (5)	State SOEs (6)	Nongovt firms (7)	State SOEs (8)
Election	.010 (.031)	-.000 (.074)	.020** (.009)	.016* (.010)	.070** (.032)	.139** (.065)	-.039 (.066)	.037*** (.008)
Close	.010 (.040)	.078 (.102)	-.014* (.008)	-.015* (.008)	-.024 (.029)			
Election X Close	.007** (.005)	.014 (.116)	.020* (.014)	.020** (.014)	.006* (.052)			
LC					-.141** (.059)	-.265* (.136)	-.005 (.009)	-.003 (.010)
Election X LC					-.105 (.085)	-.142 (.145)	-.015 (.017)	-.047 (.042)
State-level real	-.802*** (.174)	.315 (.299)	-.186*** (.049)	-.163*** (.057)	-.775*** (.199)	-.814*** (.173)	.315 (.301)	-.201*** (.055)
GDP growth	.016 (.046)	.056 (.124)	.022*** (.008)	.019* (.012)	.029 (.040)	.061* (.036)	.173* (.102)	.023*** (.008)
Constant	8.412 (.316)	8.412 (.622)	8.412 (.192)	7.017 (.178)	8.412 (.316)	8.412 (.622)	8.412 (.192)	7.017 (.240)
No. observations	8,412	8,412	8,412	7,017	8,412	8,412	8,412	7,017
R ²	.316	.622	.192	.178	.316	.622	.192	.240

This table shows the impact of political competition and patronage on corporate investments. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates come from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Election}_{it} + \beta_2 \times \text{Close}_{it} + \beta_3 \times \text{LC}_{it} + \beta_4 \times \text{Election}_{it} \times \text{Close}_{it} + \beta_5 \times \text{Election}_{it} \times \text{LC}_{it} + \beta_6 \times \text{Close}_{it} \times \text{LC}_{it} + \mu_j + \mu_t + \varepsilon_{ijt}.$$

5.5 Table 4: heterogeneity by political competition (swing targeting)

Model. This table corresponds to eq. (3) and the analogous interaction with LC.

Main message. The interaction Election $\times$ Close is positive for SOE outcomes and typically insignificant for private-firm outcomes. Hence, the SOE election-year spike is *stronger in close districts*. This aligns with tactical redistribution / swing-voter targeting.

5.6 Table 5: election-cycle dynamics

Model. This table estimates eq. (4).

Main message. For state SOEs, the coefficients on non-election years are negative: election years are the peak of the cycle. For private firms, the cycle looks different: activity is often higher mid-cycle relative to election years, consistent with election-year uncertainty or crowding-out pressures.

Table 5
Elections and corporate investment cycle

	Number of projects		Percentage
	State SOEs	Nongovt firms	<i>State Total</i>
	(1)	(2)	(3)
1 year after election	-.203*** (.064)	-.052 (.062)	-.046*** (.009)
2 years after election	-.087* (.052)	.202*** (.073)	-.021*** (.008)
2 years before election	-.138*** (.051)	-.078 (.061)	-.029*** (.009)
1 year before election	-.085** (.037)	-.030 (.076)	-.021** (.009)
State-level real GDP growth	-.778*** (.171)	.354 (.301)	-.193*** (.049)
Constant	.116*** (.027)	-.003 (.128)	.047*** (.009)
No. observations	8,412	8,412	8,412
R ²	.316	.622	.193

This table examines the temporal dynamics of corporate investments around elections. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates come from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times Sc_{-1it} + \beta_2 \times Sc_{-2it} + \beta_3 \times Sc_{-3it} + \beta_4 \times Sc_{-4it} + \gamma \times C_{it} + \mu_j + \mu_t + \epsilon_{ijt},$$

Table 6
Policy uncertainty and corporate investments

	Central SOEs			Nongovt firms	
	(1)	(2)	(3)	(4)	(5)
Policy uncertainty _t	-.035 (.040)	-.025 (.041)	-.190*** (.056)	-.598*** (.169)	-.127 (.132)
Policy uncertainty _{t-1}		.042 (.100)		-3.215*** (.418)	-.255 (.197)
Election			.212*** (.045)		-.468*** (.141)
State-level real GDP growth	.604** (.250)	.504 (.325)	.497** (.246)	5.922*** (1.262)	-1.203 (.802)
Constant	.465*** (.167)	.268 (.415)	1.074*** (.226)	5.045*** (.702)	16.480*** (1.708)
Observations	2,592	2,179	2,592	2,592	2,179
R ²	.497	.513	.505	.730	.792

This table examines the impact of policy uncertainty on corporate investments. The Policy Uncertainty index for India comes from Baker et al. (2016) and measures the average policy uncertainty in a year. This measure is only available for the period 2003–2009 and is measured at the national level necessitating the use of only national elections in these tests. The estimates come from the following panel regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \log(\text{Policy Uncertainty})_t + \gamma \times C_{it} + \mu_j + \mu_t + \epsilon_{ijt},$$

5.7 Table 6: policy uncertainty vs election timing

Model. This table implements eq. (5) in a district-FE setup.

Main message.

- Private investment is strongly negatively related to policy uncertainty (especially with lags), consistent with real-options delay under uncertainty.
- Central SOE investment is less consistently tied to policy uncertainty and remains positively linked to election years.

Interpretation. This strengthens a *political* interpretation: election-year SOE investment is not simply the mirror image of uncertainty-driven private investment.

5.8 Table 7: political authority (federal ministers)

Model. This table estimates eq. (6).

Main message. Districts represented by federal cabinet ministers receive more central SOE projects (positive Federal coefficient), consistent with politically connected allocation. The election interaction is not the sole driver: political power matters beyond the election cycle.

5.9 Table 8: political ideology (left vs right incumbents)

Model. This table estimates eq. (7) for national and state elections.

Table 7
Political authority and corporate investments

	Number of projects		Percentage
	Central SOEs	Nongovt firms	<i>Central Total</i>
	(1)	(2)	(3)
Election	.049*** (.018)	-.113 (.076)	.017** (.007)
Federal minister	.122** (.047)	-.311 (.788)	.046*** (.014)
Election X Federal minister	.109 (.086)	.272 (.387)	.028 (.023)
State-level real GDP growth	.339** (.136)	6.205*** (.962)	-.005 (.053)
Constant	.242*** (.011)	1.661*** (.124)	.069*** (.004)
Observations	5,039	5,039	5,039
R ²	.482	.552	.188

This table examines the impact of political authority of the local politician on the investments of firms around national elections. To measure political authority, we examine whether the electoral representative holds a ministerial position. The list of ministers is only available at the federal level, so these tests are only performed for national elections. The reported estimates come from the following panel regression specification:

$$Y_{ijt} = \alpha_0 + \eta Election_{it} + \phi Federal_{ijt} + \beta_1 Election_{it} \times Federal_{ijt} + \gamma \times C_{it} + \mu_j + \varepsilon_{ijt},$$

Table 8
Political ideology and corporate investments

	Number Of projects		Percentage
	Central SOEs	Nongovt firms	<i>Central Total</i>
	(1)	(2)	(3)
A. National elections			
Election	-.087*** (.031)	-.467*** (.141)	-.023* (.013)
Left	-.067*** (.022)	1.512*** (.218)	-.049*** (.009)
Election X Left	.191*** (.037)	-.168 (.131)	.061*** (.016)
State-level real GDP growth	.489*** (.133)	1.897*** (.552)	.122** (.061)
Constant	.292*** (.016)	.873*** (.137)	.099*** (.005)
Observations	5,039	5,039	5,039
R ²	.482	.562	.192
B. State elections			
Election	-.050 (.053)	-.102 (.160)	-.028* (.016)
Left	.138*** (.041)	.579*** (.168)	.019 (.012)
Election X Left	.202* (.118)	-.177 (.176)	.077*** (.020)
State-level real GDP growth	-.752*** (.259)	.738 (.463)	-.199** (.079)
Constant	.034 (.051)	-.091 (.211)	.021 (.016)
Observations	5,142	5,142	5,142
R ²	.399	.679	.232

This table examines the impact of the political ideology of the incumbent party on the investments of firms around national elections in panel A and around state elections in panel B. The reported estimates come from the following panel regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times Election_{it} + \beta_2 \times Left_{ijt} + \beta_3 \times Election_{it} \times Left_{ijt} + \gamma \times C_{it} + \mu_j + \varepsilon_{ijt},$$

Main message. The interaction Election $\times$ Left is positive for SOE investment, implying that election-year SOE manipulation is stronger (or only present) under left incumbents. This supports the view that ideological preferences mediate how governments use SOEs.

5.10 Table 9: project-level composition and visibility

Model. This table is based on eq. (8) (with IV for state elections).

Main message. Election-year SOE projects are:

- **larger** (positive Election $\times$ SOE effect on project size),
- more likely to be **expansions/new units**,
- more concentrated in **visible sectors** (construction and community development).

Private firms do not show comparable visible reallocation.

5.11 Table 10: electoral returns to SOE investments

Model. This table implements eq. (9) (within election years), and then splits visible vs other SOE projects.

Main message. More SOE projects announced in election years are associated with: (i) more constituencies won by the incumbent party, and (ii) higher incumbent vote margins. The effect is stronger for **visible** projects, consistent with political targeting for voter salience.

Table 9
Nature of election-year investments, project-level analysis

A. Project characteristics					
	Stalled/abandoned (1)	Time to completion (2)	Time to implementation (3)	Expansion (4)	Project size (5)
Election	.021*** (.007)	-.022 (.053)	.037 (.043)	.000 (.046)	.041 (.040)
SOE	-.015 (.009)	.936*** (.009)	.191*** (.009)	-.040*** (.005)	-.490*** (.052)
Election X SOE	-.018 (.014)	.111 (.135)	.074 (.084)	.015* (.077)	.263*** (.077)
Observations	17,043	9,007	12,249	17,043	13,473
Wald F-stat	42,000	42,000	42,000	42,000	42,000

B. Project industry

	Visible (1)	Construction (2)	Community development (3)	Education (4)	Health and social work (5)	Utilities (6)	Transport storage and comm. (7)	Hotels and restaurants (8)	Manufacturing (9)	Mining (10)	Real estate leasing (11)	Miscellaneous (12)
Election	.006 (.008)	.000 (.008)	.005* (.003)	-.002 (.003)	-.004 (.003)	-.013* (.007)	.001 (.005)	-.016*** (.005)	.011*** (.011)	.003 (.004)	-.007 (.008)	.001 (.002)
SOE	.288*** (.088)	.285*** (.088)	.002 (.003)	.030*** (.003)	.007*** (.003)	.087*** (.007)	.083*** (.004)	-.034*** (.005)	-.438*** (.010)	.005 (.004)	-.026*** (.007)	-.003* (.002)
Election X SOE	-.086*** (.017)	.084*** (.017)	.002 (.006)	-.013*** (.006)	.008 (.006)	.008 (.015)	-.024*** (.009)	.002 (.010)	-.021 (.022)	.001 (.008)	-.021 (.016)	-.001 (.004)
No. observations	17,043	17,043	17,043	17,043	17,043	17,043	17,043	17,043	17,043	17,043	17,043	17,043
Wald F-stat	42,000	42,000	42,000	42,000	42,000	42,000	42,000	42,000	42,000	42,000	42,000	42,000

This table examines the impact of state election cycles on project characteristics of state SOEs. These tests are only based on state elections, because the larger number of state elections allows us to use an instrumental variable (IV) estimation. The estimates are based on the following IV specification, where *Scheduled* is a dummy variable that takes a value of 1 if 5 years have passed since the previous election) is used as an instrument for *Election* and *Scheduled* X *SOE* is used as an instrument for *Election* X *SOE*:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{SOE}_{ijt} + \beta_2 \times \text{Election}_{ijt} + \beta_3 \times \text{Election}_{ijt} \times \text{SOE}_{ijt} + \mu_{ijt} + \epsilon_{ijt}$$

Table 10
Politically driven investments and election outcomes

	Constituencies won (1)	Margin of victory (2)	Constituencies won (3)	Margin of victory (4)
A. Election outcomes (SOE vs. nongovernment projects)				
Number of projects announced (SOEs)	.224*** (.051)	.009*** (.003)		
Number of projects announced (Nongovt)			.019 (.014)	.002 (.001)
Lagged margin of victory or loss	2.305*** (.654)	.087 (.055)	2.348*** (.654)	.144* (.079)
Constant	1.784*** (.303)	-.106*** (.006)	1.790*** (.303)	-.130*** (.023)
No. observations	1,747	1,747	1,747	1,747
R ²	.696	.393	.696	.516
B. Election outcomes (visible vs. other SOE projects)				
Number of projects announced (visible)	.284*** (.051)	.014*** (.007)		
Number of projects announced (others)			.099 (.116)	.003 (.004)
Lagged margin of victory or loss	2.285*** (.665)	.084 (.055)	2.165*** (.659)	.080 (.055)
Constant	1.857*** (.301)	-.105*** (.006)	1.818*** (.301)	-.102*** (.02)
No. observations	1,747	1,747	1,747	1,747
R ²	.696	.391	.688	.389

This table examines the impact of election-year SOE investments on election outcomes. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates come from the following panel regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Numproj}_{ijt} + \gamma \times C_{it} + \mu_{ijt} + \epsilon_{ijt}$$

5.12 Table 11: announcement returns (election vs non-election)

Model. This table estimates the event-study regression eq. (10) with $Z_p = \text{Election}$ for announcement returns.

Main message. The interaction $\text{Election} \times \text{SOE}$ is negative across event windows: markets react *more negatively* to election-year SOE project announcements than to comparable private announcements. This is consistent with political-value destruction (negative NPV or agency/political objectives).

5.13 Table 12: announcement returns (close vs non-close districts)

Model. This table estimates eq. (10) with $Z_p = \text{Close}$.

Main message. The interaction $\text{Close} \times \text{SOE}$ is negative: SOE projects in politically competitive districts trigger more negative market reactions. This is a sharp “political” signature: the market seems to understand that close districts induce more political distortion.

Table 11
Announcement returns of election-year SOE investments

A. Excess returns					
	Short window			Long window	
	(1)	(2)	(3)	(4)	(5)
	[-1D,+1D]	[-3D,+3D]	[-1SD,+1SD]	[-1D,+1Y]	[-1D,+3Y]
Event window					
Election	-.124 (.243)	-1.293*** (.627)	-.698 (1.488)	1.163*** (.554)	-.804 (1.651)
SOE	.577 (.611)	-.392 (2.834)	-1.843 (9.681)	-.295 (.935)	2.515 (3.645)
Election X SOE	-1.513*** (.464)	-3.739*** (1.242)	-8.089*** (2.774)	-1.906*** (.799)	-8.791*** (2.695)
Observations	2,380	2,380	2,380	2,380	2,380
R ²	.228	.195	.178	.269	.380

B. Abnormal returns

	Short window			Long window	
	[-1D,+1D]	[-3D,+3D]	[-1SD,+1SD]	[-1D,+1Y]	[-1D,+3Y]
Event window					
Election	-.171 (.373)	-1.385*** (.627)	-1.222 (1.476)	1.377*** (.550)	-.670 (1.703)
SOE	-.010 (.795)	-.915 (2.816)	-2.334 (9.620)	.085 (.919)	1.518 (3.447)
Election X SOE	-1.863*** (.421)	-2.504*** (1.163)	-5.519*** (2.301)	-1.662*** (.690)	-8.532*** (2.892)
No. observations	2,380	2,380	2,380	2,380	2,380
R ²	.237	.193	.188	.273	.358

This table presents results from tests on announcement returns of politically driven investments of central SOEs. These tests are only estimated for national elections and central SOEs, because our sample period contains only four partially privatized state-level SOEs. The estimates are based on the following specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{SOE}_{ijt} + \beta_2 \times \text{Election}_{ijt} + \beta_3 \times \text{Election}_{ijt} \times \text{SOE}_{ijt} + \beta_4 \times X_{ijt} + \mu_{ijt} + \epsilon_{ijt}$$

Table 12
Announcement returns of SOE investments in close districts

A. Excess returns					
	Short window			Long window	
	(1)	(2)	(3)	(4)	(5)
	[-1D,+1D]	[-3D,+3D]	[-1SD,+1SD]	[-1D,+1Y]	[-1D,+3Y]
Event window					
Close	.497* (.279)	.864 (.964)	3.701 (2.850)	1.407 (1.855)	2.107 (1.382)
SOE	.305 (.557)	-1.007 (2.571)	-3.039 (6.638)	-.345 (1.076)	-1.101 (3.673)
Close X SOE	-1.401*** (.464)	-3.863*** (1.242)	-9.131*** (2.523)	-2.413*** (.799)	-4.148*** (2.462)
No. observations	2,380	2,380	2,380	2,380	2,380
R ²	.227	.189	.179	.284	.370

B. Abnormal returns

	Short window			Long window	
	[-1D,+1D]	[-3D,+3D]	[-1SD,+1SD]	[-1D,+1Y]	[-1D,+3Y]
Event window					
Close	.221 (.373)	.977 (.627)	3.863 (2.871)	1.417 (1.861)	2.798* (1.610)
SOE	-.334 (.795)	-1.297 (2.567)	-2.988 (6.585)	.200 (1.045)	.641 (3.796)
Close X SOE	-1.889*** (.421)	-3.433*** (1.163)	-7.335*** (2.301)	-1.727 (.690)	-4.681*** (2.892)
No. observations	2,380	2,380	2,380	2,380	2,380
R ²	.235	.189	.189	.210	.369

This table presents results from tests on announcement returns of politically driven investments of central SOEs. These tests are only estimated for national elections and central SOEs, because our sample period contains only four partially privatized state-level SOEs. The estimates are based on the following specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{SOE}_{ijt} + \beta_2 \times \text{Close}_{ijt} + \beta_3 \times \text{Close}_{ijt} \times \text{SOE}_{ijt} + \beta_4 \times X_{ijt} + \mu_{ijt} + \epsilon_{ijt}$$

Table 13
SOE investments and private sector investments

Dependent variable	Number of nongovt projects announced				Number of nongovt projects implemented			
	L=0 (1)	L=1 (2)	L=2 (3)	L=3 (4)	L=0 (5)	L=1 (6)	L=2 (7)	L=3 (8)
Number of SOE projects	.410*** (.065)	.326*** (.052)	.323*** (.063)	.552*** (.095)	.175*** (.032)	.133*** (.026)	.154*** (.030)	.176*** (.051)
Election	-.038 (.051)	-.014 (.044)	.182*** (.054)	-.042 (.056)	-.041 (.031)	-.051* (.027)	.109*** (.032)	.000 (.033)
Election X Number of SOE projects	-.181*** (.055)	-.127*** (.043)	-.077 (.075)	-.381*** (.107)	-.075*** (.027)	-.047*** (.021)	-.042 (.045)	-.115* (.048)
State-level real GDP growth	.530* (.286)	1.063*** (.402)	1.066*** (.330)	.805*** (.357)	.276 (.185)	.773*** (.238)	.432* (.224)	.421** (.210)
Observations	8,412	7,822	7,237	6,656	8,412	7,822	7,237	6,656
R ²	.636	.662	.658	.667	.682	.708	.709	.71

This table examines the association between election-year state SOE investments and investments by the private sector. We only report these tests based on state elections as the staggered nature of these elections across time allows for cleaner identification. The estimates are based on the following specification:

$$Y_{ijt+1} = \alpha_0 + \beta_1 \times \text{Numproj}_{ijt} + \beta_2 \times \text{Election}_{ijt} + \beta_3 \times \text{Numproj}_{ijt} \times \text{Election}_{ijt} + C_{it} + \mu_{ijt} + \epsilon_{ijt}$$

5.14 Table 13: private investment crowding-out / attenuation in election years

Model. This table estimates eq. (11) for private project counts (announced/implemented), with $L = 0, 1, 2, 3$.

Main message. Outside elections, SOE and private investments are positively associated ($\beta_1 > 0$), consistent with complementarities (infrastructure spillovers). In election years, the interaction is negative ($\beta_3 < 0$): the positive association weakens, consistent with crowding out / misallocation during politically distorted periods.

6 Appendix evidence (robustness, mechanisms, and extra implications)

6.1 Table A1: monthly dynamics around state elections

Election-year effects also show up at monthly frequency: SOE project announcements rise in months leading up to the election month and drop after. This strengthens the interpretation that timing is tied to elections (not slow-moving annual shocks).

Table A1
Elections and corporate investments (monthly regressions)

State elections	Number of projects		Percentage	Cost ratio	ln(Cost)
	State SOEs (1)	Nongovt firms (2)	State SOEs (3)	State SOEs (4)	State SOEs (5)
Pre[-12 to -10]	.006** (.003)	.003 (.005)	.005** (.002)	.005** (.002)	.013 (.010)
Pre[-9 to -7]	.013*** (.003)	.012* (.006)	.010*** (.003)	.008*** (.003)	.024** (.012)
Pre[-6 to -4]	.010*** (.003)	-.002 (.006)	.010*** (.003)	.008*** (.003)	.021** (.012)
Pre[-3 to -1]	.007*** (.003)	.025*** (.007)	.004 (.003)	.003 (.003)	.014 (.012)
Post[1 to 3]	.002 (.003)	-.005 (.006)	.003 (.002)	.003 (.002)	.011 (.010)
Post[4 to 6]	.003 (.003)	.008 (.006)	.003 (.002)	.001 (.002)	-.002 (.009)
Post[7 to 9]	.004 (.003)	-.003 (.006)	.003 (.002)	.007 (.002)	.000 (.009)
Post[10 to 12]	.001 (.002)	-.011* (.006)	.001 (.002)	.001 (.002)	-.008 (.008)
State $\times$ Year FEs	Yes	Yes	Yes	Yes	Yes
Observations	106,116	106,116	106,116	103,150	103,150
R ²	.072	.287	.050	.041	.062

In this table, we reestimate the impact of state elections on corporate investments at the district-year-month level. We only report these tests based on state elections as the staggered nature of these elections across time allows us to control for state $\times$ year fixed effects. The estimates are from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \text{Post}[1, 3]_{it} + \beta_2 \text{Post}[4, 6]_{it} + \beta_3 \text{Post}[7, 9]_{it} + \beta_4 \text{Post}[10, 12]_{it} + \beta_5 \text{Post}[-12, -10]_{it} + \beta_6 \text{Post}[-9, -7]_{it} + \beta_7 \text{Post}[-6, -4]_{it} + \beta_8 \text{Post}[-3, -1]_{it} + \beta_9 \text{Post}[1, 3]_{it} + \beta_{10} \text{Post}[4, 6]_{it} + \beta_{11} \text{Post}[7, 9]_{it} + \beta_{12} \text{Post}[10, 12]_{it} + \beta_{13} \text{Post}[-12, -10]_{it} + \beta_{14} \text{Post}[-9, -7]_{it} + \beta_{15} \text{Post}[-6, -4]_{it} + \beta_{16} \text{Post}[-3, -1]_{it} + \gamma \times \text{Year FEs} + \mu_{it} + \epsilon_{ijt}$$

Table A2
Subsample robustness tests

	Off-national-cycle state elections			Scheduled state elections			Unscheduled state elections		
	State SOEs (1)	Nongovt firms (2)	State SOEs (3)	State SOEs (4)	Nongovt firms (5)	State SOEs (6)	State SOEs (7)	Nongovt firms (8)	State SOEs (9)
Election	.096*** (.033)	-.045 (.062)	.033*** (.009)	.117** (.050)	-.039 (.057)	.035*** (.008)	-.026 (.089)	-.538*** (.187)	-.005 (.020)
State-level real GDP growth	-.604*** (.179)	.416 (.285)	-.186*** (.053)	-.778*** (.195)	-.309 (.376)	-.144*** (.052)	.473 (.609)	1.309 (1.642)	-.132 (.172)
Constant	.046 (.030)	.139 (.134)	.027** (.012)	-.048 (.052)	-.121 (.134)	.012 (.009)	.159*** (.077)	1.318*** (.277)	.030 (.019)
Observations	6,177	6,177	6,177	7,181	7,181	1,231	1,231	1,231	1,231
R ²	.247	.264	.134	.265	.617	.127	.462	.736	.255

This table reports the results from our tests based on subsamples of state elections from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Election}_{it} + \gamma \times C_{it} + \mu_j + \mu_t + \epsilon_{ijt}$$

Table A3
National elections and corporate investments (publicly listed central SOEs only)

	Number Of projects		Percentage
	Central (listed) SOEs (1)	Nongovt firms (2)	Central (listed) SOEs (3)
Election	.028*** (.011)	-.003 (.029)	.013** (.005)
State level real gdp growth	.174*** (.079)	2.221*** (.337)	.030 (.045)
Constant	.058*** (.004)	.546*** (.004)	.026*** (.004)
Observations	5,039	5,039	5,039
R ²	.279	.482	.134

This table reports results of our analysis based on the subsample of investments made by publicly listed central SOEs. Because these tests are based on central SOEs, which are under the jurisdiction of the federal government, we focus on national elections. The estimates are from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Election}_{it} + \gamma \times C_{it} + \mu_j + \mu_t + \epsilon_{ijt}$$

6.2 Table A2–A3: subsamples and listed SOEs

- The state-election SOE cycle persists when excluding elections that coincide with national elections and when focusing on scheduled elections.
- Unscheduled elections show different patterns (private investment falls sharply), consistent with political crises.
- Restricting to publicly listed central SOEs still yields a positive election effect on SOE projects.

6.3 Table A4: swing targeting with state $\times$ year fixed effects

Replacing year FE with state $\times$ year FE controls for state-specific time shocks. The election $\times$ close effects for SOEs remain positive, reinforcing the targeting interpretation.

6.4 Table A5: CEO turnover

No strong evidence that CEO turnover in SOEs systematically increases in election years. This suggests politicians may influence investment decisions through channels other than leadership changes.

Table A4
Political competition, patronage, and corporate investment cycle

A. Close elections

	Close			Less contested		
	Number of projects	Percentage	Cost ratio	Number of projects	Percentage	Cost ratio
State SOEs	State SOEs	State SOEs	State SOEs	State SOEs	State SOEs	State SOEs
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Close	-.001	-.001	-.001	-.001	-.001	-.001
Electon X Close	-.001	-.001	-.001	-.001	-.001	-.001
LC	-.001	-.001	-.001	-.001	-.001	-.001
Electon X LC	-.001	-.001	-.001	-.001	-.001	-.001
State-level real GDP growth	-.001	-.001	-.001	-.001	-.001	-.001
Constant	-.001	-.001	-.001	-.001	-.001	-.001
Observations	8,412	8,412	8,412	8,412	8,412	8,412
R ²	.264	.264	.264	.264	.264	.264

This table reports the estimates from our tests examining the impact of political competition and patronage on corporate investment. We only report these tests based on state elections as the staggered nature of state elections across time allows us to control for year fixed effects. The estimates on the following regression specification:

$$Y_{ijt} = \alpha_0 + \alpha_1 \text{Electon}_{ijt} + \alpha_2 \text{LC}_{ijt} + \alpha_3 \text{Electon}_{ijt} \times \text{LC}_{ijt} + \alpha_4 \text{State-level real GDP growth}_{ijt} + \alpha_5 \text{Constant}_{ijt} + \epsilon_{ijt}$$

Table A5
Elections and CEO turnover in SOEs

	ALL central SOEs		Listed central SOEs	
	(1)	(2)	(3)	(4)
Election	-.028	-.028	-.048	-.048
4 years until next election	.019	.019	.060	.060
3 years until next election	.035	.035	.057	.057
2 years until next election	.034	.034	.078	.078
1 year until next election	.002	.002	.005	.005
Constant	.196***	.196***	.197***	.197***
Observations	3,851	3,851	374	374
R ²	.133	.134	.167	.171

This table reports the estimates from our tests examining the impact of elections on CEO turnover at SOEs. We restrict our attention to central SOEs in these tests as data on identity of CEOs is only available for these SOEs. Because these tests are based on central SOEs, which are under the jurisdiction of the federal government, we focus on national elections in these tests. The estimates are from the following regression specification:

$$Y_{ijt} = \alpha_0 + \alpha_1 \text{Election}_{ijt} + \alpha_2 \text{LC}_{ijt} + \alpha_3 \text{Electon}_{ijt} \times \text{LC}_{ijt} + \alpha_4 \text{State-level real GDP growth}_{ijt} + \alpha_5 \text{Constant}_{ijt} + \epsilon_{ijt}$$

Table A6
State elections, political competition, and corporate investment cycle

A. Close elections

	Number of projects		Percentage
	State SOEs	Nongovt firms	
(1)	(2)	(3)	
1 year after X Close	-.190*	.106	-.053**
2 years after X Close	-.167*	.107	-.027
2 years before X Close	-.238**	-.017	-.054***
1 years before X Close	-.067	-.018	-.014
Close	.226**	.206	.026*
State-level real GDP growth	-.771***	.281	-.185***
Constant	-.068	.106	.013**
Observations	8,412	8,412	8,412
R ²	.318	.623	.194
B. Less-contested elections			
1 year after X LC	.210**	.018	.051**
2 years after X LC	.131	.151	.001
2 years before X LC	-.090	.092	.013
1 year before X LC	.015	.039	-.014
LC	-.083	.179	.019
State-level real GDP growth	-.760***	.294	-.185***
Constant	.203***	.119	.052***
Observations	8,412	8,412	8,412
R ²	.264	.593	.130

This table reports the estimates from our tests examining temporal dynamics of investments in closely contested (less-contested) districts in panel A (B) and around elections. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates come from the following regression specification:

$$Y_{ijt} = \alpha_0 + \alpha_1 \text{Electon}_{ijt} + \alpha_2 \text{LC}_{ijt} + \alpha_3 \text{Electon}_{ijt} \times \text{LC}_{ijt} + \alpha_4 \text{State-level real GDP growth}_{ijt} + \alpha_5 \text{Constant}_{ijt} + \epsilon_{ijt}$$

6.5 Table A6: cycle dynamics interacted with political competition

Close districts display stronger election-cycle dynamics for SOE investments, consistent with tactical manipulation.

6.6 Table A7: visible vs other projects at the district level

Election effects concentrate in *visible* SOE projects and in the visible share of SOE spending.

Table A6
State elections, political competition, and corporate investment cycle

A. Close elections

	Number of projects		Percentage
	State SOEs	Nongovt firms	
(1)	(2)	(3)	
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1 years before X Close	-.067	-.018	-.014
Close	.226**	.206	.026*
State-level real GDP growth	-.771***	.281	-.185***
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Observations	8,412	8,412	8,412
R ²	.318	.623	.194
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1 year after X LC	.210**	.018	.051**
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This table reports the estimates from our tests examining temporal dynamics of investments in closely contested (less-contested) districts in panel A (B) and around elections. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates come from the following regression specification:

$$Y_{ijt} = \alpha_0 + \alpha_1 \text{Electon}_{ijt} + \alpha_2 \text{LC}_{ijt} + \alpha_3 \text{Electon}_{ijt} \times \text{LC}_{ijt} + \alpha_4 \text{State-level real GDP growth}_{ijt} + \alpha_5 \text{Constant}_{ijt} + \epsilon_{ijt}$$

Table A7
Changing the composition of election-year investments, district-level analysis

	Number Of projects		Percentage		Visible		Cost ratio
	Visible	Other	State Total	Visible	Other	Visible cost	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	
Election	.089**	.017	.016***	.041**	.053	.023***	
State-level real	-.371***	-.435***	-.403**	-.443***	-.469**	-.097*	
GDP growth	-.130	.110	.044	.136	.271	.041	
Constant	-.017	.037*	.004	-.028*	.084*	-.010**	
Observations	33,099	33,099	33,099	26,000	26,000	26,000	

The empirical setup in this table is the same as that in panel C of Table 3. This table reports the estimates from tests based on the sample of state elections from the instrumental variables regression, where *Scheduled* (a dummy variable that takes the value of 1 if 5 years have passed since the previous election) is used as an instrument for *Election*. *Visible projects* comprise the SOE projects announced in the Construction and Community Development (public and recreation centers) industry sectors. The dependent variable *Y* refers to one of the following variables: the *Number of projects* announced in a district by state SOEs that are in visible and those in other industries in Columns 1 and 2, respectively; *Percentage*, the ratio of total number of visible investment projects announced by state SOEs to total number of all investment projects in a district in Column 3; *ln(Cost)*, the natural log of one plus the total costs of visible and other investments by state SOEs in Columns 4 and 5, respectively; and *Cost ratio*, the ratio of total costs of visible investment projects by state SOEs to total costs of all investment projects by state SOEs in Column 6. *Election* is a dummy variable that identifies election years. The coefficient on *Scheduled* in the first-stage instrumental variable regression (not reported here for brevity) is .95. The last row reports Cragg-Donald Wald F-statistics for weak identification from the first stage.

Table A8
Political competition, politically driven investments, and state election outcomes

	Constituencies won		Margin of victory	
	(1)	(2)	(3)	(4)
Number of projects announced (SOE)	.191***	.009**		
Number of projects announced (nongovt)	-.048	.015	.002	
Close	-.300*	-.018	-.027	-.025**
Number of projects announced (SOE) X Close	.074	.002	.189	.012
Number of projects announced (nongovt) X Close	.056	.009	.010	.001
Lagged margin of victory or loss	1.946***	.062	1.957**	.108
Constant	1.907***	-.098***	1.907***	-.119***
Observations	1,747	1,747	1,747	1,747
R ²	.697	.393	.697	.518

This table reports the estimates from our tests examining the impact of election year SOE investments in closely contested districts on election outcomes. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates are from the following regression specification:

$$Y_{ijt} = \alpha_0 + \alpha_1 \text{Electon}_{ijt} + \alpha_2 \text{Close}_{ijt} + \alpha_3 \text{Electon}_{ijt} \times \text{Close}_{ijt} + \alpha_4 \text{Lagged margin of victory}_{ijt} + \alpha_5 \text{Constant}_{ijt} + \epsilon_{ijt}$$

where *i* refers to the state, *j* refers to the district, and *t* refers to the year. The dependent variable *Y* is *Constituencies won*, the number of constituencies in a district where the winner belonged to the incumbent's party during the current elections in Columns 1 and 3, and *Margin of victory*, the difference in share of votes received by incumbent and opposition parties in a district in the current election in Columns 2 and 4. The appendix defines all variables. These tests are based on State elections. The election data come from the election commission of India, and data on new project announcements come from CAPEX, a database of new projects announced in India. We control for district and year fixed effects in these tests. In these tests, standard errors are robust to heteroscedasticity and clustered at the district level. **p* < .1; ***p* < .05; ****p* < .01.

6.7 Table A8: election outcomes and political competition

SOE projects predict better incumbent outcomes; interactions with close districts are not strongly different, suggesting visible SOE activity has broad electoral effects.

6.8 Table A9: firm-level investment regressions (Prowess)

Firm-level data confirm that central SOEs increase investment measures in national election years, while private firms do not.

Table A9
National Elections and Firm Level Investment Regressions

	Central SOEs			Nongovt firms		
	(1) <i>Investments</i> Total assets	(2) # projects	(3) ln(Project value)	(4) <i>Investments</i> Total assets	(5) # projects	(6) ln(Project value)
Election	.027*** (.007)	.410** (.155)	.562** (.279)	-.031 (.031)	-.010 (.006)	-.004 (.014)
Size _{t-1}	-.017 (.012)	.470* (.253)	.917** (.433)	-.097 (.153)	.089*** (.009)	.184*** (.017)
Tobins' Q _{t-1}	-.001 (.002)	.021 (.083)	.144 (.113)	1.051*** (.270)	.001 (.001)	.002 (.002)
Cash _{t-1}	.001 (.006)	.070 (.305)	-.089 (.365)	.001 (.002)	.000 (.000)	.000 (.001)
State-level real GDP growth _{t-1}	-.056 (.068)	2.469 (3.141)	6.644** (3.216)	-.471 (.685)	.603*** (.065)	1.260*** (.139)
Constant	.185 (.119)	-3.596 (2.448)	-7.439* (4.231)	.343 (.915)	-.456*** (.056)	-.904*** (.110)
Observations	574	574	574	37,613	37,613	37,613
R ²	.096	.614	.578	.776	.277	.230

This table reports the estimates from our tests examining the impact of election on capital expenditure at the firm level. We have to restrict our attention to central SOEs in these tests as data on firm-level capital expenditure is only available for these SOEs. Because these tests are based on central SOEs, which are under the jurisdiction of the Federal government, we focus on national elections in these tests. The estimates are from the following regression specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Election}_{it} + \gamma \times C_{it} + \mu_i + \varepsilon_{ijt}$$

Where i refers to firm, and t refers to year. The dependent variable Y is gross investments scaled by beginning of the year total assets made by firm i in year t in Columns 1 and 4, the number of projects announced by a firm in a year in Columns 2 and 5, and natural log of the total value of all projects announced in Columns 3 and 6. The data cover the period 1995–2009. The election data come from the election commission of India, and data on capital expenditures come from Prowess, a database that provides financial information for firms in India. C_{it} refers to firm level controls that include lagged values of firm size, sales growth, cash flow, and Tobin's q . We also control for lagged values of state level real GDP growth and firm fixed effects in these tests. Standard errors are robust to heteroscedasticity and clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.9 Table A10: univariate announcement-return patterns

Univariate comparisons show that election-year SOE projects (and projects in close districts) have strongly negative announcement returns, while private projects do not show the same negative pattern.

6.10 Table A11–A12: implementation returns

Implementation-date returns show weaker patterns than announcement-date returns, consistent with information being incorporated at the announcement stage (implementation is less “surpris-

Table A10

Univariate Tests: Announcement Returns of Politically Driven Investments

Variable	Election=1 (1)	Election=0 (2)	Difference (3)	Close=1 (4)	Close=0 (5)	Difference (6)
A. Excess returns						
Central SOE	Excess returns SE	-1.098*** (.375)	.369 (.280)	-1.468*** (.461)	.143 (.499)	-1.469*** (.532)
	N	139	217	84	272	
Nongovt	Excess returns SE	-.063 (.189)	.444*** (.124)	-.507** (.228)	.559** (.226)	.218* (.218)
	N	745	1,787	571	961	.341 (.249)
B. Abnormal returns						
Central SOE	Abnormal returns SE	-1.396*** (.386)	.332 (.328)	-1.728*** (.512)	-1.337** (.539)	-.079 (.287)
	N	139	217	84	272	
Nongovt	Abnormal returns SE	.184 (.210)	.506*** (.131)	-.322 (.244)	.877*** (.249)	.279** (.124)
	N	745	1,787	571	961	.597** (.269)

This table presents results from our univariate tests on announcement returns of politically driven investments of central SOEs. These tests are based on central SOEs and national elections because there are only four partially privatized state-level SOEs during our sample period. Announcement return is defined as *Excess return (Abnormal Return)* on the firm's stock over the day of the project announcement in panel A (panel B). * $p < .1$; ** $p < .05$; *** $p < .01$.

Table A11

Implementation Returns of Election Year SOE Investments

Event Window	Short window			Longer window	
	(1) [-1D,+1D]	(2) [-3D,+3D]	(3) [-15D,+15D]	(4) [-1D,+1Y]	(5) [-1D,+3Y]
A. Excess returns					
Election	-.055 (.293)	.314 (.654)	-.150 (1.240)	.120 (.573)	-1.509 (1.649)
SOE	.675 (.805)	.766 (1.397)	-.820 (3.057)	-.820 (1.282)	5.576** (2.252)
Election X SOE	-1.811*** (.602)	-.435 (1.583)	-.567 (2.228)	-.848 (.884)	-3.718 (2.844)
Observations	1,820	1,820	1,820	1,820	1,820
R ²	.252	.282	.250	.347	.456
B. Abnormal returns					
Election	.241 (.389)	.623 (.692)	-.131 (1.259)	.178 (.574)	-1.501 (1.649)
SOE	.147 (.758)	.636 (1.472)	-1.932 (3.065)	-.911 (1.285)	5.117** (2.179)
Election X SOE	-.326 (1.008)	-.359 (1.778)	-.121 (2.273)	-.868 (.889)	-3.561 (2.849)
Observations	1,820	1,820	1,820	1,820	1,820
R ²	.309	.284	.260	.345	.455

This table presents results from our multivariate tests on implementation returns of politically driven investments of central SOEs. These tests are based on central SOEs and national elections because there are only four partially privatized state-level SOEs during our sample period. The estimates are from the following specification:

$$Y_{fpt} = \alpha_0 + \beta_1 \times (\text{SOE}_f) + \beta_2 \times \text{Election}_{pt} + \beta_3 \times \text{Election}_{pt} \times (\text{SOE}_f) + \beta_4 \times X_f + \mu_{f,ind} + \varepsilon_{fpt}$$

Where f refers to firm and p refers to project. The dependent variable Y_{fp} is *Excess return (Abnormal return)* on the firm's stock around the date of the project implementation in panel A (panel B). The unit of observation is a firm's project. *Election* is a dummy variable that takes the value of 1 for projects announced in election years. *SOE* is a dummy variable that takes the value of 1 for central SOEs and 0 for nongovernment firms. We control for *Industry fixed effects* and lagged value of other firm-specific variables: *ROA*, *Debt/Assets*, *Size*, *Tobin's q* in all of these tests. The appendix defines all variables. Standard errors are robust to heteroscedasticity and clustered at the firm level. * $p > .1$; ** $p > .05$; *** $p > .01$.

ing”).

Table A12
Implementation Returns of SOE Investments in Close Districts

Event window	Short window			Longer window	
	(1)	(2)	(3)	(4)	(5)
	[-1D,+1D]	[-3D,+3D]	[-15D,+15D]	[-1D,+1Y]	[-1D,+3Y]
<i>A. Excess returns</i>					
Close	.437 (.323)	-.404 (.530)	1.078 (1.391)	-.368 (.702)	3.226* (1.829)
SOE	.291 (.731)	.494 (1.184)	-2.232 (2.905)	-2.769 (2.184)	5.267** (2.433)
Close X SOE	-1.684** (.744)	.726 (1.304)	-3.579 (2.585)	-.817 (1.681)	-6.453 (4.035)
Observations	1,820	1,820	1,820	1,820	1,820
R ²	.250	.282	.251	.297	.538
<i>B. Abnormal returns</i>					
Close	-.668** (.327)	-.643 (.494)	.449 (1.399)	-.370 (.704)	3.221* (1.833)
SOE	-.123 (.642)	.264 (1.160)	-1.879 (3.122)	-2.873 (2.213)	4.823** (2.329)
Close X SOE	.882 (.813)	1.588 (1.328)	-.612 (2.945)	-.902 (1.695)	-6.182 (3.996)
Observations	1,820	1,820	1,820	1,820	1,820
R ²	.312	.284	.260	.296	.537

This table presents results from our multivariate tests on implementation returns of politically driven investments of central SOEs. These tests are based on central SOEs and national elections because there are only four partially privatized state-level SOEs during our sample period. The estimates are from the following specification:

$$Y_{fpt} = \alpha_0 + \beta_1 \times (\text{SOE}_f) + \beta_2 \times \text{Election}_{pt} + \beta_3 \times \text{Election}_{pt} \times (\text{SOE}_f) + \beta_4 \times X_f + \mu_{f,ind} + \epsilon_{fpt}$$

Where f refers to firm and p refers to project. The dependent variable Y_{fpt} is *Excess return (Abnormal return)* on the firm's stock around the date of the project implementation in panel A (panel B). The unit of observation is a firm's project. *Close* is a dummy variable that takes the value of 1 for districts if the Incumbent party's Margin of victory or loss in the district is less than 5% and 0 otherwise. *SOE* is a dummy variable that takes the value of 1 for central SOEs and 0 for nongovernment firms. We control for *Industry fixed effects* and lagged value of other firm-specific variables: *ROA*, *Debt/Assets*, *Size*, *Tobin's q* in all of these tests. The appendix defines all variables. Standard errors are robust to heteroscedasticity and clustered at the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table A13
State elections, SOE investments, and crowding out of nongovernment investments (cost of projects)

Dependent Variable	Total cost of nongovt projects announced			
	L=0	L=1	L=2	L=3
Year	(1)	(2)	(3)	(4)
<i>A.</i>				
Number of SOE projects	.120*** (.030)	.090*** (.029)	.094*** (.036)	.201*** (.039)
Election	-.007 (.052)	-.047 (.051)	.118** (.057)	-.007 (.056)
Election X Number of SOE projects	-.059** (.024)	-.048* (.026)	-.063 (.040)	-.156*** (.046)
State-level real GDP growth	.060 (.378)	1.002** (.412)	.658 (.404)	.722 (.452)
Constant	.712*** (.090)	1.666*** (.083)	1.202*** (.071)	.737*** (.073)
Observations	7,017	6,590	6,196	5,802
R ²	.543	.557	.56	.561

This table presents results from our multivariate tests examining whether election year state SOE investments crowds out investments by the private sector. These tests are based on state elections as the staggered nature of these elections across time allows us to control for year fixed effects. The estimates are from the following specification:

$$Y_{ijt} = \alpha_0 + \beta_1 \times \text{Numpoj}_{ijt} + \beta_2 \times \text{Election}_{it} + \beta_3 \times \text{Numpoj}_{ijt} \times \text{Election}_{it} + \gamma \times C_{it} + \mu_j + \mu_t + \epsilon_{ijt}$$

6.11 Table A13: crowding out using private-project costs

Using $\log(1+\text{cost})$ of private projects yields the same qualitative result: the positive association between SOE and private investment weakens in election years.

6.12 Figures A1–A3: visual summaries

- **Figure A1/A2:** monthly event-time plots show a clear pre-election spike in state SOE projects and spending, not mirrored by private firms.
- **Figure A3:** cumulative excess returns are negative for election-year SOE investments and positive for private investments, matching the event-study regressions.

7 Short summary

1. The paper provides micro evidence consistent with the **political view of state ownership**: SOEs are used as instruments of political strategy.
2. The evidence is not just timing: it is stronger in **competitive districts** and shifts toward **visible** projects; it also maps into **electoral outcomes**.
3. The market evidence (negative announcement returns) suggests **value destruction** from politically driven projects at partially privatized SOEs.
4. Real allocation effects appear via **attenuation/crowding out** of private investment linkages during election years.

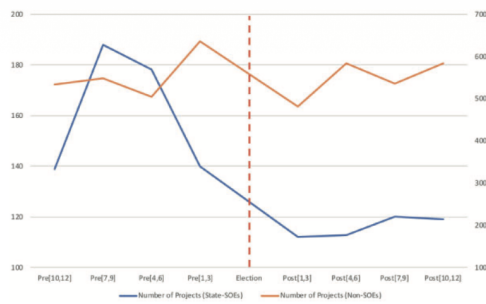


Figure A1
Time series of number of projects
 This figure plots the number of projects announced by state SOEs and nongovernment firms around the month of election. *Post[1,3]* is a dummy variable that takes the value one for months 1, 2, and 3 after the month of election. *Post[4,6]*, *Post[7,9]*, and *Post[10,12]* are defined analogously. *Pre[1,3]* is a dummy variable that takes the value one for months 1, 2, and 3 before the month of election. *Pre[4,6]*, *Pre[7,9]*, and *Pre[10,12]* are defined analogously. Dashed lines represent months that coincide with state elections.

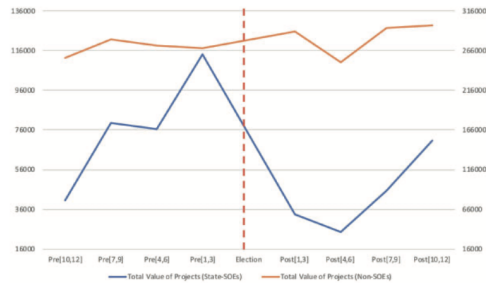


Figure A2
Time Series of value of projects
 This figure plots the total value of projects (in INR millions) announced by state SOEs and nongovernment firms around the month of election. *Post[1,3]* is a dummy variable that takes the value one for months 1, 2, and 3 after the month of election. *Post[4,6]*, *Post[7,9]*, and *Post[10,12]* are defined analogously. *Pre[1,3]* is a dummy variable that takes the value one for months 1, 2, and 3 before the month of election. *Pre[4,6]*, *Pre[7,9]*, and *Pre[10,12]* are defined analogously. Dashed lines represent months that coincide with state elections.

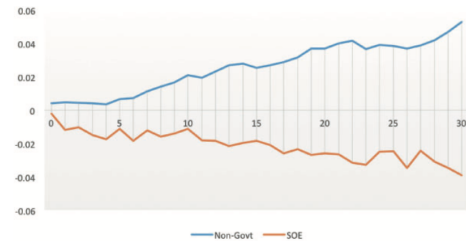


Figure A3
Cumulative excess return of election year corporate investments
 This figure shows cumulative return over 30 days for election-year projects announced by central SOEs and nongovernment firms.

8 Conclusion

8.1 Summary

This paper provides micro-level evidence that **state ownership transmits electoral incentives into corporate capital allocation**. Exploiting India’s constitutionally mandated election schedule (and the predictable five-year cycle) as a source of plausibly exogenous variation in politicians’ incentives to cater to voters, the authors show that **state-owned enterprises (SOEs) systematically expand and re-orient investment around elections**. Using project-level data on capital expenditure announcements (CAPEX, 1995–2009), they find a **political investment cycle** in SOE capital spending that is (i) stronger in politically salient places and (ii) targeted toward **visible infrastructure-type projects**.

The main economic implication is a clear **politics–value tradeoff**: election-year SOE project announcements help incumbents electorally, but markets view these investments as value-reducing for shareholders, consistent with politically motivated (negative NPV) “pork-barrel” investment.

8.2 Main empirical conclusions

1. **Election years shift SOE investment upward (project counts).** Controlling for district and year fixed effects, the number of projects announced by SOEs rises in election years. The paper summarizes the magnitude as roughly

$$\text{State SOEs: } +27.5\%, \quad \text{Central SOEs: } +17\%.$$

Nongovernment firms do *not* display a comparable election-year spike.

2. **Election years change the *composition* of SOE investment toward “visible” spending.** SOEs disproportionately increase investment in sectors such as construction/community development and other infrastructure-oriented categories in election years, consistent with a strategy of delivering (or credibly promising) benefits that voters can easily observe prior to an election (a “visibility” channel).
3. **Election-year SOE projects are larger.** Conditional on being announced, SOE projects in election years are about **17% larger** than SOE projects in off-election years (while there is no comparable election-year deterioration in completion time or abandonment probability in the descriptive evidence).
4. **Political targeting: effects are stronger when electoral incentives are stronger.** The election-year SOE investment response is amplified in districts with *close elections*, in districts represented by *high-ranking politicians*, and is driven especially by *left-leaning incumbents*. These patterns support an incentives-based interpretation rather than a purely macro/uncertainty story.

5. **Electoral returns: SOE projects help incumbents.** Election outcomes improve with SOE project announcements. In the election-outcome regressions (Table 10), the coefficient on SOE projects implies:

each additional SOE project $\Rightarrow$ about 0.9% higher incumbent margin of victory ($\hat{\beta} \approx 0.009$).

Visible SOE projects have an even stronger association with electoral success (e.g., $\hat{\beta} \approx 0.014$ for margin of victory), whereas nongovernment projects do not show comparable effects.

6. **Shareholder-value costs: markets react negatively to politically driven SOE investments.** Using event-study windows around project announcements for partially privatized *central* SOEs (Table 11), the interaction term is strongly negative:

$$\hat{\beta}_{\text{Election} \times \text{SOE}} \approx -1.5\% \text{ } ([-1, +1] \text{ days}) \quad \text{to} \quad \approx -8.1\% \text{ } ([-15, +15] \text{ days})$$

for excess returns, with similar results using CAPM abnormal returns. Projects announced by SOEs in *close districts* also have more negative announcement returns (Table 12).

7. **Back-of-the-envelope welfare cost is economically large.** Translating announcement-return losses into market-cap losses, the paper reports a rough shareholder deadweight cost for three national elections (1999, 2004, 2009) ranging from about

$$\text{\$2.4B--\$13B} \quad (\text{using } -1.5\%) \quad \text{to} \quad \text{\$13B--\$69B} \quad (\text{using } -8.0\%).$$

The paper emphasizes these are likely lower bounds (political influence can operate beyond projects and beyond election years).

8. **Spillovers exist but are muted in election years.** SOE investment is positively associated with private-sector investment on average (a “crowding-in” pattern), but this association is significantly *reduced* in election years (Table 13), consistent with election-year SOE projects being less efficiency-enhancing and/or more distortionary.

8.3 Economic intuition: why the results fit together

The results are coherent under a simple political-economy mechanism:

- **SOEs are controllable policy instruments.** Relative to private firms, SOEs can be pressured to undertake investments that serve political objectives (employment, local visibility, electoral support) even when such projects do not maximize shareholder value.
- **Elections raise the marginal value of visible investment.** When reelection is at stake, politicians prefer projects whose benefits are salient and quickly attributable (roads, construction, community development), because they shift voter beliefs about competence and perfor-

mance. This yields a predictable *political investment cycle* in both levels and composition of SOE capex.

- **Close races intensify distortions.** In competitive districts the electoral return to additional spending is higher, so the same SOE-capex “lever” is pulled more aggressively, creating stronger cycles and more negative market reactions.
- **Shareholders pay the price.** Public shareholders of partially privatized SOEs bear the cost of politically motivated investments, which shows up as negative announcement returns and large implied deadweight losses.

8.4 Takeaway

India’s election cycle causally distorts SOE investment: SOEs announce more (and more visible) projects in election years, which improves incumbents’ electoral outcomes but reduces shareholder value, supporting the political view of state ownership.